Mengzhen Li

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Education

University of Minnesota, Minneapolis, MN Aug. 2021 – Anticipated May 2027
M.S./Ph.D. Computer Science

College of Saint Benedict, Saint Joseph, MN Jan. 2018 – May 2021
B.A. Computer Science

Technical Skills

Languages: Java, Python, C++, SQL, SML
Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Android Studio, PyCharm, IntelliJ
Operating Systems: Linux, Windows, Mac OS, Ubuntu

Experience

Industry Research Collaborator June 2025 – June 2026
Medtronic and University of Minnesota Collaboration

- Completed a Medtronic-sponsored project designing **automated log analysis** and **sequence-based anomaly detection** for large-scale, real-world datasets to improve failure management and accelerate root-cause analysis.

Graduate Research Assistant Aug. 2021 – Present
University of Minnesota *Minneapolis, MN*

- Built **ARUS**, which automatically **generated behavior-preserving fixes** for Java test code by analyzing runtime executions and safely removing unnecessary stubbings in Java tests; improved reliability across 128 open-source projects. Contributed to open-source communities (e.g., Google-oauth, LDBC); **85 PRs merged**.
- Developed **REMI**, an automated tool to **detect and fix mismatched invocations** in Java test suites. Evaluated on 217 open-source projects; fixed **171/172** issues with minimal changes; submitted patches (**7 merged**).
- Built **REDUST** to automatically reduce duplicated test code; across 213 Java projects, it converted **204 duplicated configurations into 68 reusable test helpers** while preserving software behavior and reducing maintenance complexity.

Undergraduate Research Assistant May 2020 – Aug. 2021
College of Saint Benedict *Saint Joseph, MN*

- Designed **SynpleTest**, an educational tool based on **program synthesis**, automatically generating test scenarios to help beginners reason about test adequacy and design tradeoffs.
- Conducted user studies showing over **64% usability approval**, validating synthesis-driven feedback for learning testing complexity.

Publications

Mengzhen Li, Mattia Fazzini. "Automatically Removing Unnecessary Stubbings from Test Suites." *IEEE ICST 2024 (Distinguished Paper Award).*

Mengzhen Li, Mattia Fazzini. "Automatically Resolving Mismatched Invocations on Test Doubles in Test Suites." *IEEE ICST 2026.*

Jamie Lee, Flynn Teh, Hengcheng Zhu, **Mengzhen Li**, Mattia Fazzini, Valerio Terragni. "Improving LLM-Driven Test Generation by Learning from Mocking Information." *IEEE ICSTW 2026.*

Mengzhen Li, Mattia Fazzini. "DataTD: A Dataset of Java Projects Including Test Doubles." *IEEE/ACM MSR 2025.*

Alessio Gambi, **Mengzhen Li**, Mattia Fazzini. "Artisan: An Action-Based Test Carving Tool for Android Apps." *IEEE ICSME 2023.*

Mengzhen Li, Mattia Fazzini. "Automatically Refactoring Duplicated Stubbing Configurations in Test Suites." *Under submission.*

Awards and Honors

ICST 2024 Distinguished Paper Award 2024
IEEE International Conference on Software Testing, Verification and Validation (ICST)